

Soft Computing (IE422)

Instructor: Pushpendra Kumar

Course Placement: *Soft Computing* (IE422) is an ICT & Technical Elective / MnC Elective.

Course Format: 3 Lecture hours and 2 Lab hours for the 4-credit course.

Course Prerequisites: Basic knowledge of mathematics fundamentals (including discrete mathematics), and MATLAB/python.

Course Content: This course is designed for UG students, aiming to build a strong foundation as well as advanced understanding of Soft Computing techniques. It starts with the basics of soft computing, its characteristics, and how it differs from hard computing. The fuzzy logic part covers fuzzy sets, relations, membership functions, fuzzy inference rules, fuzzification and defuzzification methods along with their applications. The neural network section introduces artificial neuron models, activation functions, perceptrons, backpropagation, associative memory, radial basis function networks, and fuzzy neural networks. The genetic algorithm module explains the GA framework, encoding, selection, crossover, mutation, optimization problems, convergence theory, and fuzzy logic controlled genetic algorithms. The course balances theoretical aspects with practical applications, enabling students to apply these techniques in real-life and engineering problem-solving. Overall, the content equips learners with essential skills in fuzzy logic, neural networks, and genetic algorithms, making it relevant for research as well as practical implementation.

Text Books:

- 1.) Timothy J. Ross, *Fuzzy Logic with Engineering Applications* (John Wiley & Sons, 2009)
- 2.) S. Haykin, *Neural Networks and Learning Machines*, third Edition, New Delhi, India: PHI Learning, 2011.
- 3.) S. N. Sivanandam and S. N. Deepa, *Principles of Soft Computing* (John Wiley & Sons, 2007).
- 4.) D. E. Goldberg, *Genetic Algorithms in Search, Optimization and Machine Learning*, Pearson Education, 2006.

Reference Books:

- 1.) S. Rajasekaran and G. A. Vijayalakshmi Pai, *Neural Networks, Fuzzy Systems and Evolutionary Algorithms: Synthesis and Applications*, second Edition, New Delhi, India: PHI Learning, 2017.
- 2.) J. S. R. Jang, C. T. Sun, and E. Mizutani, *Neuro-Fuzzy and Soft Computing* (Prentice Hall, 1997).

Assessment Method: The total grade will be based on lab assignments (30%) and scheduled examinations (two in-semesters and one end-semester) (70%).

Course Outcomes:

This course will enable students to understand the fundamental principles of fuzzy logic, neural networks, and genetic algorithms, and apply soft computing methods in real-world applications such as data analysis, control, and decision-making. Learners will gain both theoretical insights and practical problem-solving skills in soft computing techniques. The course provides a strong foundation for advanced research or industry roles in artificial intelligence, machine learning, and computational intelligence. After successful completion of the course, the student will have the ability to:

CO1: Apply fuzzy logic, neural networks, and genetic algorithms to optimization and decision-making problems.

CO2: Recognize the feasibility of using soft computing methods for solving real-world and engineering applications.

CO:3 Demonstrate analytical and computational skills in designing intelligent systems using soft computing approaches.

POs-COs Matrix:

P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12
X	X	X		X							

TENTATIVE LECTURE SCHEDULE

Sl. No.	DESCRIPTION		No. of Lectures
1	Introduction to Soft Computing		4
	1.1	Fundamentals of soft computing, Soft vs. Hard computing	
	1.2	Key features, advantages, and domains of application	
2	Fuzzy Sets and Fuzzy Logic		17
	2.1	Basics of classical vs. fuzzy sets and relations	
	2.2	Representations of fuzzy sets, properties, and set operations	
	2.3	Cardinality, operations, fuzzy cartesian product and composition, fuzzy tolerance and equivalence relations, forms of composition operation	
	2.4	Fuzzy numbers, linguistic variables, arithmetic operations	
	2.5	Membership functions, fuzzy to crisp conversions	
	2.6	Classical vs. Fuzzy logic, approximate reasoning	
	2.7	Concepts such as support, α -cut, height, and projections	
	2.8	Linguistic hedges, and rule-based fuzzy inference systems, canonical forms	
	2.9	Implication methods, fuzzification–defuzzification techniques, and real-world uses	
	2.10	Membership value assignments: intuition, inference, rank ordering	
3	Neural Networks		10
	3.1	Introduction to artificial neural networks	
	3.2	Activation functions, learning paradigms, and associative memory	
	3.3	Hopfield and BAM neural networks	
	3.4	Perceptron models: single-layer and multi-layer structures	
	3.5	Radial basis function networks and applications	
	3.6	Fuzzy neural networks, generation of membership functions using a neural network	
4	Genetic Algorithms		9
	4.1	Principles of genetic search, GA framework, and architectural variants	
	4.2	GA operators: encoding, crossover, selection, mutation	
	4.3	Application of GA to single-objective optimization	
	4.4	Convergence analysis and performance considerations	
	4.5	Computing membership functions using genetic algorithms	
	4.6	Fuzzy logic controlled genetic algorithms	

LAB TOPICS

1. **Introduction to MATLAB/Python for Soft Computing**
 - Basic operations, visualization, and toolboxes for fuzzy, neural networks, and optimization.
2. **Fuzzy Sets and Membership Functions**
 - Implement triangular, trapezoidal, Gaussian membership functions and visualize them.
3. **Fuzzy Inference Systems**
 - Design a simple rule-based fuzzy system (e.g., temperature control, speed control).
4. **Defuzzification Methods**
 - Implement Centroid, Bisector, MOM, SOM, LOM techniques with examples.
5. **Membership value assignments**
 - Intuition, inference, rank ordering
6. **Neural Network Basics**
 - Implementation of single-layer perceptron and testing with linear separable data.
7. **Multi-Layer Perceptron (MLP) with Backpropagation**
 - Train an MLP for classification (e.g., XOR problem, digit recognition).
8. **Radial Basis Function Networks**
 - Design and compare with MLP on approximation/classification tasks.
9. **Hopfield and BAM neural networks**
 - Qualitative and quantitative analysis
10. **Fuzzy neural networks, Generation of membership functions using NNs**
 - Qualitative and quantitative analysis
11. **Genetic Algorithm (GA) Operators**
 - Implement encoding, selection, crossover, mutation on a sample problem.
12. **GA for Optimization**
 - Apply GA to solve unconstrained single-objective optimization (e.g., minimizing a nonlinear function).
13. **Integration of all Techniques**
 - Computing membership functions using genetic algorithms
 - Fuzzy Logic Controlled Genetic Algorithms